

WESTHAMPTON GABRESKI, NY, USA

WMO: 744865

Lat: 40.844N

Lon: 72.632W

Elev: 20

StdP: 101.08

Time Zone: -5.00 (NAE)

Period: 98-19

WBAN: 14719

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.7	-11.1	-20.9	0.6	-11.0	-18.6	0.7	-9.1	12.9	-3.9	11.9	-2.9	2.5	340	0.527

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.8	30.9	22.6	29.0	21.9	27.6	21.6	24.5	28.0	23.9	26.9	23.2	25.9	4.6	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.5	18.3	26.1	22.9	17.6	25.4	22.4	17.1	25.0	74.1	27.9	71.5	27.0	69.0	25.8	27.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.0	9.4	8.4	-18.7	34.2	3.1	2.1	-20.9	35.7	-22.7	36.9	-24.5	38.1	-26.7	39.6		
			-19.1	25.9	3.0	0.9	-21.3	26.5	-23.1	27.0	-24.8	27.5	-27.0	28.1		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.4	-1.2	-0.4	3.0	8.2	13.4	18.8	22.3	21.8	18.1	12.0	6.7	2.0
	DBStd	9.21	5.56	4.67	4.48	3.85	3.72	3.42	2.91	2.94	3.64	4.45	4.58	5.04
	HDD10.0	1355	349	293	221	79	11	0	0	0	0	30	121	253
	HDD18.3	3226	607	526	477	306	158	36	4	5	49	202	351	507
	CDD10.0	1512	1	1	3	24	117	263	382	365	242	91	20	4
	CDD18.3	339	0	0	0	1	6	49	127	111	41	5	0	0
	CDH23.3	2394	0	0	0	9	48	326	964	839	196	12	0	0
	CDH26.7	559	0	0	0	2	7	79	262	183	24	1	0	0
Wind	WSAvg	3.8	4.4	4.3	4.5	4.1	3.6	3.3	3.2	2.9	3.2	3.7	3.9	4.1
Precipitation	PrecAvg	1110	86	77	113	96	85	87	80	92	105	101	82	113
	PrecMax	1471	170	136	313	178	169	214	146	236	202	476	146	202
	PrecMin	797	43	26	16	39	13	2	21	20	26	6	10	37
	PrecStd	161	34	31	66	41	38	59	32	54	45	89	35	40
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.2	13.1	17.9	24.1	27.6	32.1	34.1	32.3	29.1	25.0	18.8	16.2
		MCWB	11.6	8.3	11.8	14.2	18.8	22.5	23.9	23.3	22.2	19.4	14.9	13.4
	2%	DB	11.7	10.7	14.0	19.4	24.2	28.9	31.4	30.1	27.0	22.3	17.4	14.0
		MCWB	10.5	8.3	8.7	12.2	17.2	20.9	22.7	22.4	21.1	18.8	14.7	12.5
	5%	DB	9.7	9.0	11.9	17.0	22.2	26.9	29.2	28.6	25.5	21.0	16.2	12.6
		MCWB	7.8	6.9	8.5	11.0	16.5	20.1	22.1	22.1	20.5	17.8	13.8	11.2
	10%	DB	7.8	7.3	10.1	14.8	20.0	24.9	27.7	27.4	24.1	19.5	14.8	10.9
		MCWB	5.8	5.0	7.3	9.9	15.0	19.3	21.8	22.0	20.1	16.8	12.6	9.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.2	10.8	13.4	17.1	20.7	23.8	25.6	25.6	24.1	21.2	17.1	14.7
		MCDB	12.8	11.6	16.2	21.7	25.4	28.6	31.3	29.5	26.9	23.1	18.0	15.4
	2%	WB	10.5	8.9	10.7	14.1	19.1	22.6	24.5	24.7	23.2	19.9	15.7	13.0
		MCDB	11.5	10.2	12.3	17.0	22.8	26.8	28.5	27.7	25.1	21.5	16.9	13.7
	5%	WB	8.1	7.4	9.1	12.6	17.7	21.6	23.8	24.0	22.3	18.7	14.4	11.5
		MCDB	9.3	8.7	11.1	14.9	20.6	24.9	27.2	26.7	24.1	20.5	15.8	12.4
	10%	WB	6.2	5.2	7.7	11.4	16.4	20.7	23.1	23.4	21.3	17.4	12.9	9.2
		MCDB	7.6	7.0	9.6	13.6	19.1	23.4	26.1	25.8	23.3	19.0	14.5	10.6
Mean Daily Temperature Range	5% DB	MDBR	9.7	10.3	10.1	10.9	10.5	9.9	9.8	9.9	10.4	10.9	10.6	9.9
		MCDBR	10.9	11.8	12.4	14.4	13.4	12.5	11.7	10.9	10.7	11.2	10.5	10.0
		MCWBR	10.2	9.9	9.0	8.6	7.6	6.4	5.2	5.5	6.6	8.1	8.7	9.9
	5% WB	MCDBR	10.4	10.8	10.7	10.5	10.7	9.6	8.7	7.8	7.6	8.6	9.0	9.4
		MCWBR	10.2	10.1	9.0	7.4	6.6	5.5	4.5	4.5	5.4	7.2	8.4	9.9
Clear-Sky Solar Irradiance	taub	0.311	0.320	0.340	0.384	0.435	0.474	0.490	0.464	0.395	0.350	0.320	0.311	
	taud	2.509	2.484	2.470	2.367	2.259	2.193	2.155	2.225	2.386	2.519	2.555	2.552	
	Ebn at Noon	851	896	914	889	847	811	793	803	842	850	836	824	
	Edh at Noon	74	88	99	118	135	144	148	135	107	84	71	67	
All-Sky Solar Radiation	RadAvg	1.83	2.71	3.76	4.87	5.45	5.95	6.01	5.41	4.40	3.00	2.07	1.55	
	RadStd	0.15	0.26	0.31	0.50	0.53	0.58	0.32	0.34	0.34	0.27	0.17	0.13	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)	
(47)	Regional (64 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+106	+59	(47)	

Nomenclature: See separate page